

Assignment 1: Food Delivery Time Analysis

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Data Source

Food Delivery Time Prediction Data set — 50,000 simulated food delivery orders with 24 variables.

1. Import the Dataset

```
delivery <- read.csv("Food_Delivery_Time_Prediction.csv")
str(delivery)
```

```
## 'data.frame':    50000 obs. of  24 variables:
## $ Order_ID       : chr  "ORD000001" "ORD000002" "ORD000003" "ORD000004" ...
## $ Order_Date     : chr  "2025-10-28" "2025-09-26" "2025-01-26" "2025-11-19" ...
## $ Order_Hour     : int   8 12 18 12 18 18 12 22 8 16 ...
## $ Day_of_Week    : chr   "Tuesday" "Friday" "Sunday" "Wednesday" ...
## $ Is_Weekend     : int   0 0 1 0 0 0 0 0 0 0 ...
## $ Is_Festival    : int   0 0 0 0 0 0 0 0 1 0 ...
## $ Weather        : chr   "Clear" "Rain" "Clear" "Cloudy" ...
## $ Pickup_Zone    : chr   "CBD" "CBD" "Residential" "Residential" ...
## $ Dropoff_Zone   : chr   "Commercial" "CBD" "Residential" "Industrial" ...
## $ Vehicle_Type   : chr   "Scooter" "Scooter" "Scooter" "Bike" ...
## $ Rider_Experience_Years : num  3.1 1.8 3.8 1.7 12.4 1.6 2.2 0.7 10.2 9.3 ...
## $ Rider_Rating   : num  3.8 3.6 3.9 3.9 4.4 3.5 4 3.7 4.4 4.3 ...
## $ Restaurant_Rating : num  3.8 3.9 4 4.6 4 4.9 4.3 3.6 3.6 3.4 ...
## $ Cuisine_Type   : chr   "Pizza" "Desserts" "Biryani" "Burger" ...
## $ Order_Items    : int   1 3 1 3 1 1 5 3 3 6 ...
```

```
## $ Restaurant_Load      : chr  "Medium" "Medium" "Low" "Medium" ...
## $ Preparation_Time_Min : int   23 11 31 19 7 16 13 23 33 18 ...
## $ Road_Distance_km    : num   30.55 2.25 10.15 26.33 30.48 ...
## $ Delivery_Distance_Category: chr   "Long" "Short" "Long" "Long" ...
## $ Traffic_Level        : chr   "Moderate" "Moderate" "Moderate" "Low" ...
## $ Number_of_Signals    : int   15 17 8 9 8 16 10 14 17 10 ...
## $ Average_Speed_kmph   : num   28.7 23.3 28.9 41.4 31.7 24.2 13.4 29.3 29.2 45.9 ...
## $ Delivery_Priority    : chr   "Normal" "Normal" "Normal" "Normal" ...
## $ Time_taken_min       : int   91 21 52 59 75 105 180 37 110 69 ...
```

The dataset contains 50,000 rows and 24 columns, with no missing values in any column. It includes both quantitative variables (delivery time, road distance, preparation time) and categorical variables (weather, traffic level, vehicle type).

2. Descriptive Statistics — Quantitative Variables

Three quantitative variables were selected: delivery time, road distance, and preparation time.

```
summary(delivery[, c("Time_taken_min", "Road_Distance_km", "Preparation_Time_Min")])
```

```
## Time_taken_min Road_Distance_km Preparation_Time_Min
## Min.      : 10.00   Min.      : 0.08   Min.      : 5.00
## 1st Qu.: 58.00   1st Qu.:16.33   1st Qu.:18.00
## Median : 78.00   Median :25.62   Median :24.00
## Mean   : 83.87   Mean   :26.20   Mean   :24.03
## 3rd Qu.:102.00   3rd Qu.:35.48   3rd Qu.:30.00
## Max.    :180.00   Max.    :70.82   Max.    :51.00
```

Delivery time averages roughly 84 minutes with a median around 78. The mean sitting above the median suggests the distribution is right-skewed, which is confirmed in the histogram below.

3. Descriptive Statistics — Categorical Variables

Two categorical variables were selected: traffic level and weather.

```
table(delivery$Traffic_Level)
```

```
##
##      High      Low Moderate   Severe
##      3744     32567     13165      524
```

```
table(delivery$Weather)
```

```
##
## Clear Cloudy   Fog   Rain   Storm
## 23992 11085   3413  9073   2437
```

```
# As percentages
round(100 * prop.table(table(delivery$Traffic_Level)), 1)
```

```
##
##      High      Low Moderate      Severe
##      7.5      65.1      26.3      1.0
```

Most deliveries occur under low-to-moderate traffic and clear weather, with “Severe” traffic and “Storm” conditions being comparatively rare categories.

A useful cross-check is comparing the quantitative variable across these categorical groups:

```
round(tapply(delivery$Time_taken_min, delivery$Traffic_Level, mean), 1)
```

```
##      High      Low Moderate      Severe
##     111.7     76.4     92.3     134.4
```

Average delivery time rises steadily with traffic level, from roughly 76 minutes under low traffic to over 130 minutes under severe traffic.

4. Variable Transformations

Transformation 1 — log transform (quantitative). Delivery time is right-skewed, so a natural-log transformation was applied to compress the long upper tail.

```
delivery$log_time <- log(delivery$Time_taken_min)
head(delivery[, c("Time_taken_min", "log_time")])
```

```
##      Time_taken_min log_time
## 1              91 4.510860
## 2              21 3.044522
## 3              52 3.951244
## 4              59 4.077537
## 5              75 4.317488
## 6             105 4.653960
```

Transformation 2 — level collapsing (categorical). The five-level Weather variable was collapsed into a simpler two-level indicator of adverse versus clear conditions.

```
delivery$adverse_weather <- factor(
  ifelse(delivery$Weather %in% c("Rain", "Storm", "Fog"), "Adverse", "Clear/Cloudy")
)
table(delivery$adverse_weather)
```

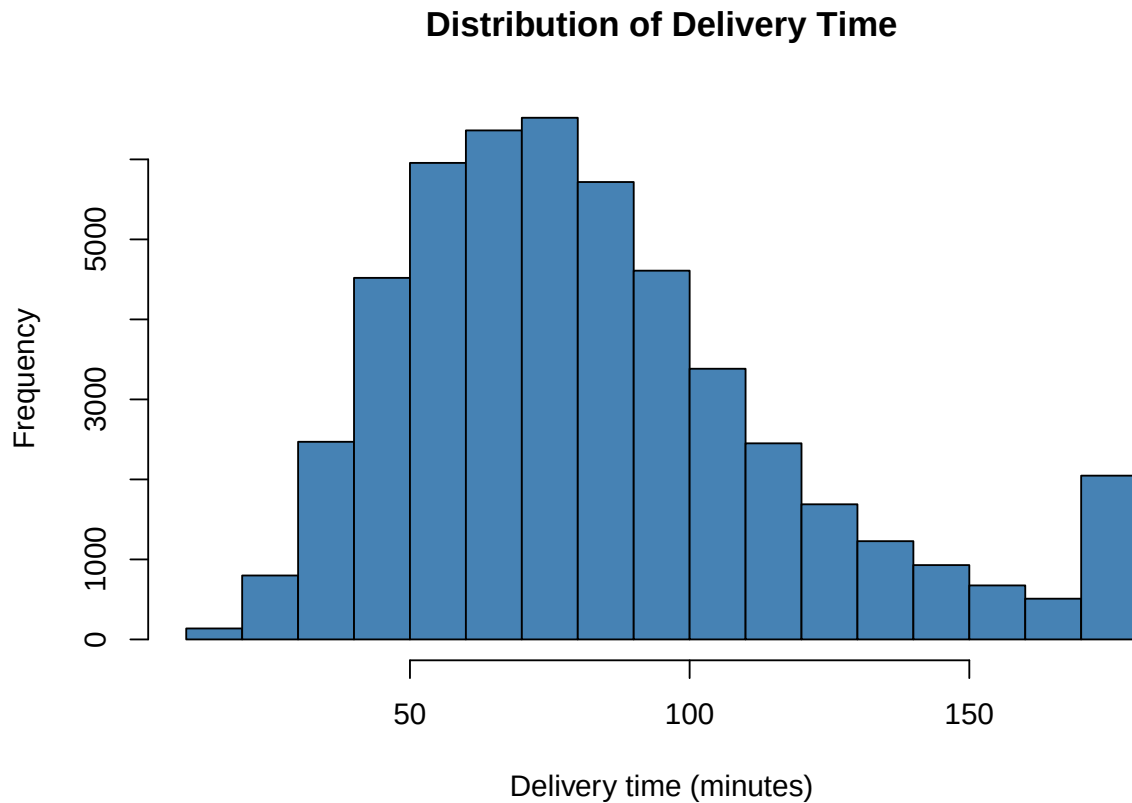
```
##
##      Adverse Clear/Cloudy
##     14923      35077
```

About 30% of deliveries (14,923 of 50,000) occurred in adverse weather.

5. Plots

Histogram of a quantitative variable.

```
hist(delivery$Time_taken_min,
     main = "Distribution of Delivery Time",
     xlab = "Delivery time (minutes)",
     col = "steelblue")
```

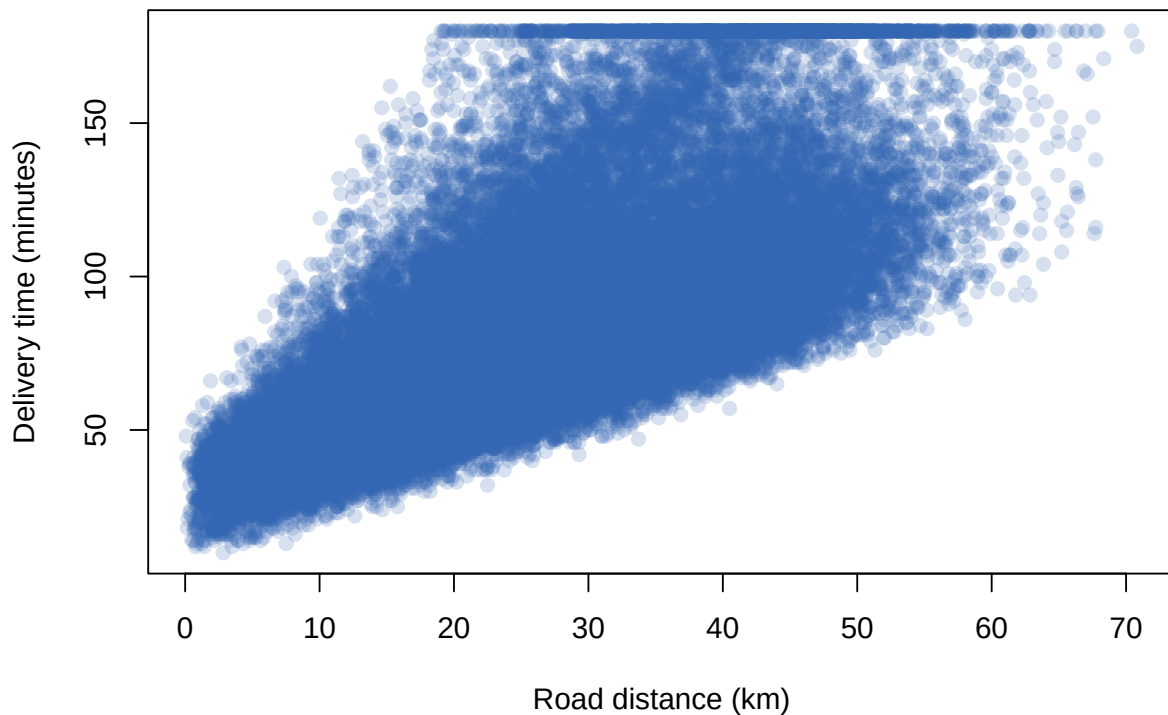


The histogram confirms the right skew noted earlier: most deliveries cluster between roughly 60 and 90 minutes, with a tail extending out toward the 180-minute ceiling.

Scatterplot.

```
plot(delivery$Road_Distance_km, delivery$Time_taken_min,
     main = "Delivery Time vs. Road Distance",
     xlab = "Road distance (km)",
     ylab = "Delivery time (minutes)",
     pch = 19,
     col = rgb(0.2, 0.4, 0.7, 0.2))
```

Delivery Time vs. Road Distance



There is a clear positive relationship between road distance and delivery time.. Farther deliveries take longer, as expected. The flat band of points along the top of the plot (delivery time capped near 180 minutes) is visible here as well. Overall, it reflects a ceiling in how the data was generated rather than a naturally occurring pattern in delivery logistics.

Summary

This analysis imported a 50,000-row food delivery data set, summarized three quantitative and two categorical variables, applied a numeric (log) and a categorical (level-collapsing) transformation, and produced a histogram and a scatter plot. The clearest finding is that both traffic level and distance are strongly associated with delivery time, while the 180-minute cap in the data is a generation artifact worth noting rather than a real-world limit.